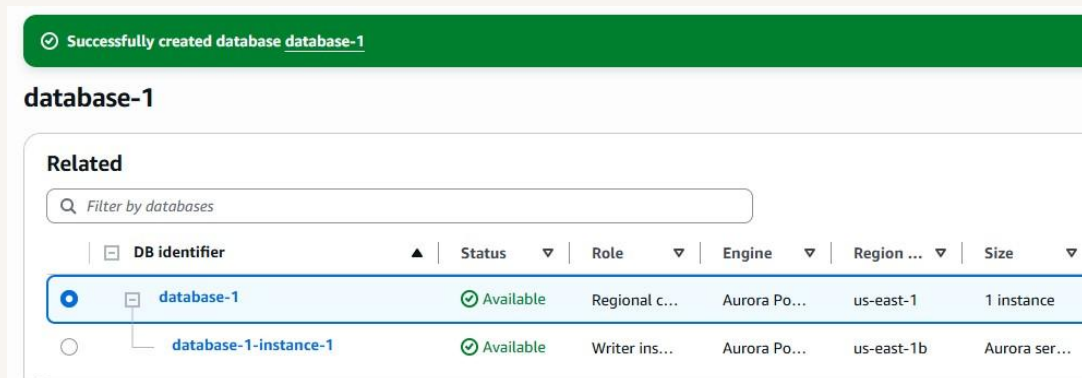


Connect a Web App to Amazon Aurora

Melvin green



Introducing Today's Project!

What is Amazon Aurora?

Amazon Aurora is a type of relational database, and it's useful because I need something large-scale, with strong peak performance and uptime.

How I used Amazon Aurora in this project

In today's project, I used Amazon Aurora to create an Aurora MySQL Database.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was the time it takes for the database to be created. Normally, most AWS products are quick to create.

This project took me...

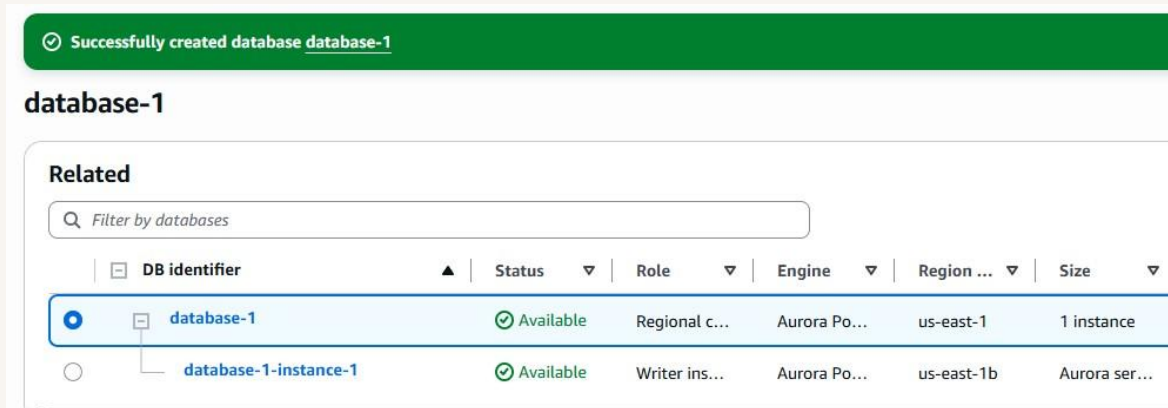
This project roughly took me 45 mins to complete.

In the first part of my project...

Creating an Aurora Cluster

A relational database is a type of database that organizes data into tables, which are collections of rows and columns, kind of like a spreadsheet.

Aurora is a good choice for large-scale applications that need strong peak performance and high uptime. This is because Aurora databases use clusters.



Halfway through I stopped!

I paused creating my Aurora database because I first need to set up an EC2 instance for it to connect to.

Features of my EC2 instance

I created a new key pair for my EC2 instance so that I can securely SSH into it.

When I created my EC2 instance, I took particular note of the public IPv4 DNS and the value of the key pair name. The IPv4 DNS is the location of the virtual machine, while the key pair name is the key needed to get into the virtual machine.

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12...[read more](#)

ami-028ba4d4ccb4b7b72

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

 Preview code

Then I could finish setting up my database

Connectivity

Info



Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource

Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ Connect to an EC2 compute resource

Set up a connection to an EC2 compute resource for this database.

EC2 instance

Info

Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

i-0915ceb717c82e9fa

ec2-instance-web-server

▼



 Some VPC settings can't be changed when a compute resource is added

Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group `rds-ec2-X` is added to the database and another called `ec2-rds-X` to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.

Aurora databases use clusters because a cluster is a group of database copies that work together, so my data is always available. Each cluster consists of a primary instance (where all write operations occur) and multiple read replicas as backups. If the primary instance fails, one of the replicas can be promoted to primary automatically.